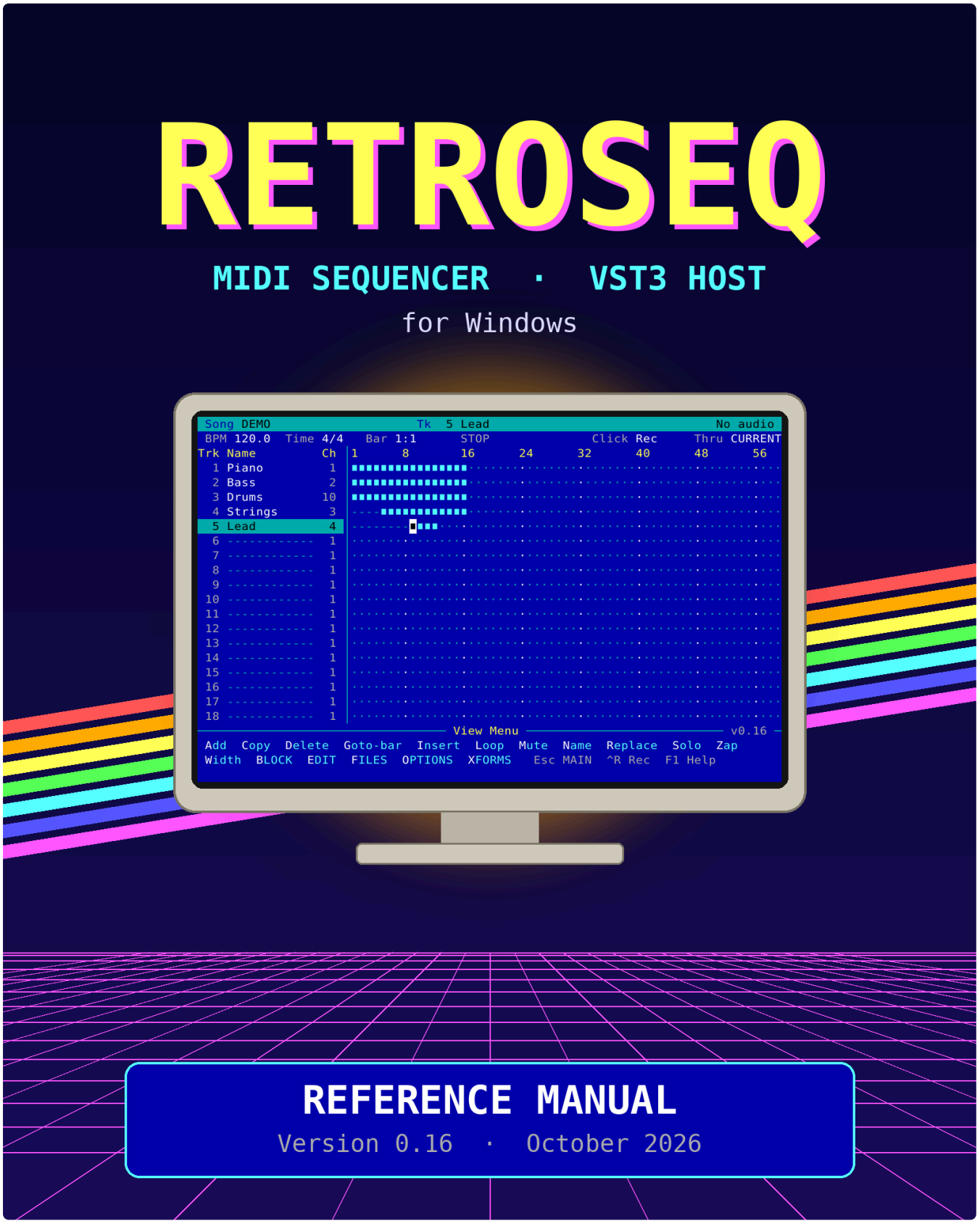
RetroSeq Reference Manual

Oct 25, 2026 · @Michael Östman



RetroSeq is a keyboard-driven MIDI sequencer and VST3 host for Windows, modelled on the classic DOS sequencers of the late 1980s. This manual covers version 0.16 and follows the layout of a classic sequencer reference manual: getting started, the sequencer screens one by one, then the global functions and a quick reference card.

Table of Contents

Getting Started

1. Introduction · System Requirements · About this Manual
2. The RetroSeq User Interface — screens, function keys, Ctrl keys, keyboard commands, using a mouse
3. Setting Up the PC MIDI System — building, audio and MIDI devices, ASIO, VST3 instruments
4. MIDI Basics — messages, channels and ports, General MIDI, timing
5. Basic Concepts — screens, building a song, controlling the MIDI network, recording techniques
6. Tutorial — record a first song, arrange with buffers, correct notes

Sequencer

7. Main Screen — status area, work area, column examples, menu commands, record and playback
8. View Screen — work area, cursor keys, menu commands, buffers, track length, block moves, record and playback from any bar, punch-in
9. Edit Screen — work area, time units, moving the cursor, menu commands, manipulating notes, step record, MIDI Edit
10. XFORMS — time, velocity, pitch, length and split transforms

Global Functions

11. Global Functions — Files (export and import a track), Options, Configuration, Metronome, MIDI Thru, Play Range, MIDI clock out
12. Troubleshooting — no sound, timing, hanging notes, ports and plug-ins

Appendix

13. Quick Reference Card · File Formats · Not Yet Implemented · Version History · Glossary · Index

1. Getting Started

Introduction

RetroSeq brings the fast, keyboard-only way of working of the classic DOS sequencers to a modern Windows PC. Where the originals talked to an MPU-401 card and external synths, RetroSeq uses the Windows MIDI ports and can also play VST3 instruments on any track. Songs are standard MIDI files, so they open in any other sequencer.

Everything is done from the keyboard. Each screen has a menu at the bottom; press the first letter of a command to run it. Function keys and Ctrl keys work the same on every screen.

System Requirements

Item	Requirement
Operating system	Windows 10 or 11, 64-bit
Audio	Any Windows audio device (WASAPI or DirectSound) or an ASIO driver
MIDI	Any MIDI interface or USB MIDI device recognised by Windows (optional)
Instruments	External MIDI synths and/or VST3 instruments
To build from source	Visual Studio 2022 or 2026 with *Desktop development with C++*, Git

An audio device must be open even if you only play external MIDI synths: the audio card drives the sequencer's clock.

About this Manual

The chapters follow the order of the screens: Main, View and Edit, then the windows that are available everywhere. Keys are written in brackets, for example **[F3]** or **[Ctrl]-O**. A menu command is written with its first letter in bold, for example **R**ecord, because that letter runs it.

The examples use a small demo song with five tracks: Piano, Bass, Drums, Strings and Lead, 16 bars at 120 BPM in 4/4.

2. The RetroSeq User Interface

Every screen has the same three areas, from top to bottom: the **status area** (song, current track, tempo, position, mode), the **work area** (what the screen is about) and the **menu area** (the commands of that screen, with the version number on its top line). The bottom line shows messages and prompts.

In the menu area, commands are written with a capital first letter (**D**elete, **L**oop) and other screens or windows in capitals (**VIEW**, **OPTIONS**). Press the first letter to run either. **[Esc]** goes back to the previous screen, closes a window or cancels a command.

Function Keys

These work on every screen.

Key	Function	With [Shift]
[F1]	Help (function key summary)	QWERTY synth (not yet)
[F2]	Metronome on/off	Metronome options
[F3]	Options window	—
[F4]	Configuration: audio/MIDI devices, VST3 scan	Markers (not yet)
[F5]	Play Range: play a bar range once or in a loop	Program names (not yet)
[F8]	Current bar (go to a bar)	Solo current track
[F9]	Chase (not yet)	MIDI Thru OFF / CURRENT
[F10]	Quit	—

Ctrl Key Combinations

Keys	Function
[Ctrl]-B	Backup: restore the current track as it was before the last recording or edit; again = redo
[Ctrl]-C	Set MIDI channel
[Ctrl]-E	Open the VST3 instrument's own editor window (Main and View)
[Ctrl]-G, [Ctrl]-J	Go to track
[Ctrl]-O	Set output: a MIDI port or a VST3 instrument

Keys	Function
[Ctrl]-P	Set program number (on a VST3 track: choose from the plug-in's program list)
[Ctrl]-R	Record (in View: from the cursor)
[Alt]-P	Panic: all notes off on every output

Keyboard Commands

Key	Function
First letter	Runs a menu command
[Enter]	Executes a command or confirms a value
[Esc]	Back to the previous screen, close a window, cancel
[Spacebar]	Play and stop; starts recording when Record is armed
[Tab], [Shift]-[Tab]	Main and View: next/previous track with data. Edit: next/previous note
+ −	Change a value by one
[ ]	Change a value by ten (transpose and pitch: by an octave)
< >	Edit screen: change start or length by single clicks
[Ins], [Del]	View: insert/remove a bar. Edit: insert/delete a note

Using a Mouse

The mouse works as in the classic sequencers, with the extras you expect in Windows (version 0.16 and later). Everything can still be done from the keyboard.

Action	Effect
Click a menu word (Delete, VIEW, XFORMS …) or a hint (Esc, F1, ^R)	Runs that command
Middle button, or left and right together	Mouse menu with the screen's commands; click an item, double-click to run it

Action	Effect
Right double-click	[Esc]
Main: click a cell	Selects the track and column
Main: drag up or down on a value	Changes it: left button ±1, right button ±10
Main: double-click	On a cell = [Enter] on it; on the track number = View screen
View: click, or drag along a track	Moves the cursor, or marks bars; then C, D, Z or X works on them
View: double-click	Edit screen on that bar
Edit: double-click	Inserts a note, or deletes the note under the pointer
Edit: drag a note	Moves it; time snaps to Units, up and down changes the pitch
Edit: drag a note's last column	Changes its length
MIDI Edit: click a field, then drag or turn the wheel	Selects and changes the value
Wheel	Main: changes the value under the pointer ([Ctrl] = ×10); elsewhere scrolls tracks, pitches or events; [Shift] scrolls bars or time
Lists and Options	Click selects, double-click confirms, the wheel scrolls or changes

In Block Moves, drag the rectangle with the mouse and double-click instead of [Enter].

Example: to give the bass more level, point at its Vol cell on the Main screen, hold the right button and drag up two steps: Vol 90 becomes 110.

3. Setting Up the PC MIDI System

Installing RetroSeq

Run `RetroSeq-<version>-Setup.exe` and follow the wizard. It asks whether to install for all users (Program Files, needs administrator rights) or only for you. No other software is needed: RetroSeq.exe carries its own C++ runtime.

Installed	Where
RetroSeq.exe	Program folder; Start menu → RetroSeq; optional desktop icon
This manual as PDF	`Manual` in the program folder; Start menu → RetroSeq Reference Manual
Example songs	`Examples` in the program folder; Start menu → Example Songs
Open with for .mid files	Optional: right-click a MIDI file → Open with → RetroSeq

The example songs are DEMO.mid (used throughout this manual), Waltz in 3-4.mid, Drum Loop - 4 bars.mid and Bass Line - 4 bars.mid (made for FILES → Import), and Straight Hats - try swing.mid (for Super-Quantize swing). They have no outputs: choose one per track with [Ctrl]-O. Save your own songs somewhere else, for example in Documents.

Uninstall from Windows Settings → Apps or the Start menu. Settings, the VST3 plug-in list and MIDI clock ports stay in `%APPDATA%\RetroSeq` so a new version starts where you left off.

To build the installer yourself, see `Installer\` in the source folder: build x64-Release, install Inno Setup (`winget install JRSoftware.InnoSetup`) and run `build-installer.cmd`.

Building RetroSeq

RetroSeq is delivered as source code and built with Visual Studio. The first build downloads the JUCE framework from GitHub, so it needs an internet connection and takes a few minutes.

1. Open Visual Studio and choose *File → Open → Folder…*. Pick the folder that contains `CMakeLists.txt`.
2. Wait until the Output window says *CMake generation finished*.
3. Choose the **x64-Release** configuration and **RetroSeq.exe** as the startup item.
4. Choose *Build → Build All*, then run with **[Ctrl]+[F5]**.

Keep the build folder out of cloud-synced folders such as Dropbox; it holds several hundred megabytes of intermediate files.

Audio and MIDI Devices

Press **[F4]** and choose *Audio and MIDI devices*.

- **Audio device type**: *Windows Audio* (WASAPI) works everywhere; choose *ASIO* if your interface has an ASIO driver, for the lowest latency.
- **Output**: the pair of outputs you listen to. With a mixer interface, pick the pair routed to the main mix.
- **Buffer size**: 480 or 512 samples (about 10 ms) is a good start. Raise it if heavy VST3 instruments crackle.
- **Active MIDI inputs**: tick the keyboards you play. Leave mixer, DAW-control and virtual editor ports unticked; every ticked input is recorded.

VST3 Instruments

Press **[F4]**, choose *VST3 plug-ins (scan)*, then *Options → Scan for new or updated VST3 plug-ins*. The standard folder `C:\Program Files\Common Files\VST3` is scanned by default; add other folders with **+**. Large instruments can take a while to load during the first scan.

If a plug-in crashes RetroSeq during a scan, start RetroSeq and scan again: the plug-in that was being scanned is skipped and listed as failed.

Testing the System

1. On the Main screen, move to track 1 and press **[Ctrl]-O**. Choose a MIDI output or a VST3 instrument.
2. Play your keyboard. The track under the cursor always receives MIDI in (MIDI Thru CURRENT), so you hear it at once.
3. Press **[Spacebar]** to play the song, and again to stop.

4. MIDI Basics

MIDI sends instructions, not sound: which key went down, how hard, and when it went up. A synth or a VST3 instrument turns those instructions into sound. This chapter explains the few MIDI ideas you need to use RetroSeq well.

Messages

Message	What it says	Where you meet it in RetroSeq
Note on	Key number 0–127 (60 = middle C) and velocity 1–127	Every recorded note; the Edit screen

Message	What it says	Where you meet it in RetroSeq
Note off	The key was released	The end of a note in the Edit screen
Program change	Choose sound 0–127	The Prg column
Control change 7	Channel volume 0–127	The Vol column on MIDI tracks
Control change 10	Pan 0–127, 64 = centre	The Pan column on MIDI tracks
Control change 64	Sustain pedal down/up	Recorded from your pedal; [Alt]-P sends it up
Pitch bend, aftertouch, other CCs	Wheels, pressure, knobs	Recorded and played back as they are

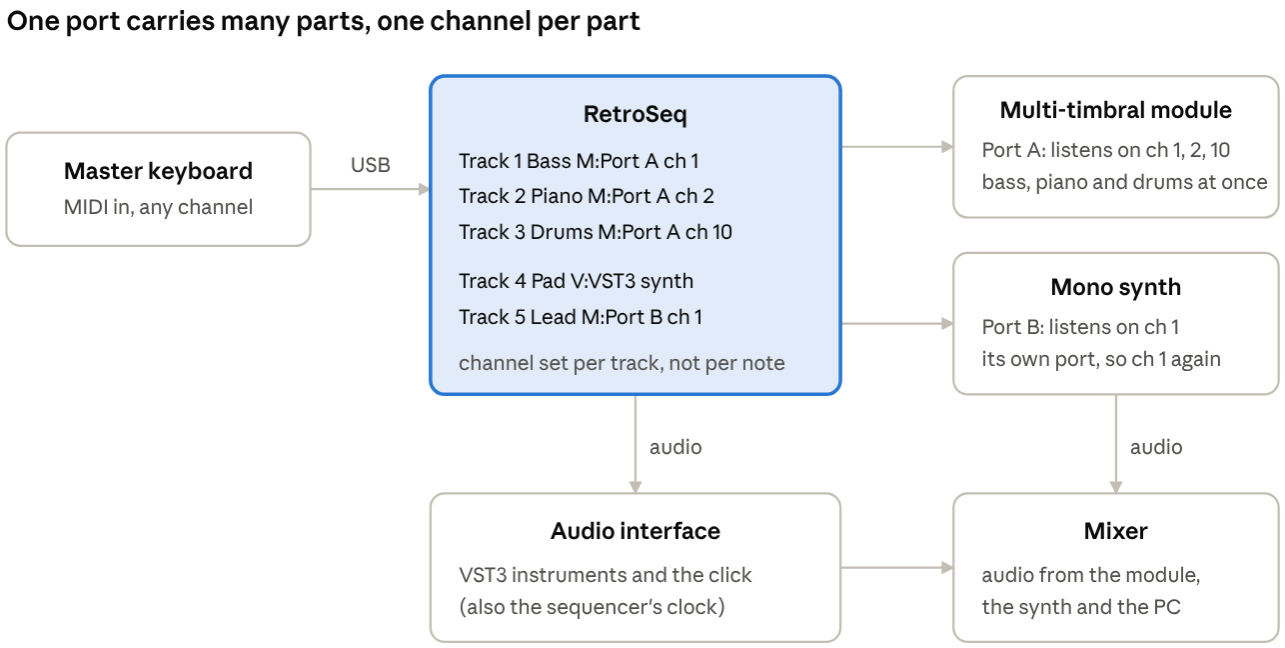
Velocity is how hard a key was struck. Most sounds play louder and brighter with higher velocity. Velocity scaling in the Vol column adds or subtracts a fixed amount from every note.

Channels and Ports

Every MIDI message carries a **channel**, 1–16. A synth set to receive on channel 2 ignores everything on other channels, so one cable can carry sixteen separate parts. A **multi-timbral** synth plays a different sound on each channel at once.

A **port** is one MIDI output: a socket on an interface or a USB connection to a synth. Each port has its own sixteen channels.

In RetroSeq the channel belongs to the track: whatever channel you play on, the track sends on its **Ch** setting. By General MIDI convention, drums are on channel 10.



a small MIDI network · 2 ports, 4 MIDI parts, 1 VST3 instrument

Track 5 can use channel 1 again because it plays through its own port. The VST3 instrument on track 4 needs no channel at all: it is the only thing listening to that track.

General MIDI

General MIDI (GM) is a standard set of 128 sounds with fixed program numbers and a fixed drum map on channel 10, so a song sounds roughly right on any GM module. In GM, program 0 is a grand piano, 32 an acoustic bass and 48 a string ensemble; on the drum channel, note 36 is the bass drum, 38 the snare and 42 a closed hi-hat. The demo song uses that drum map.

Timing

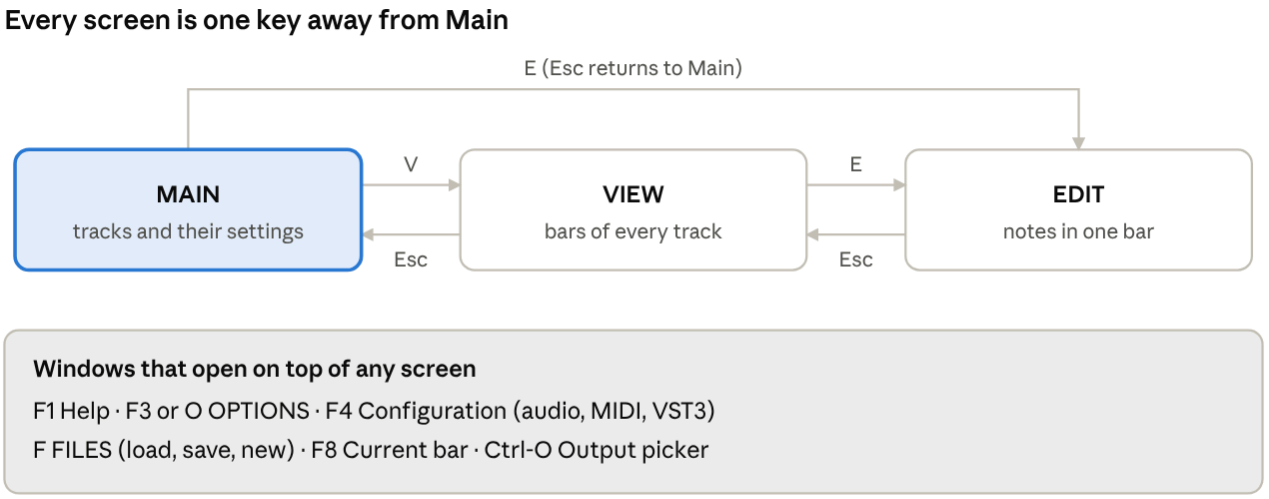
RetroSeq stores time in clicks: 480 per quarter note. A bar of 4/4 is 1,920 clicks, a sixteenth note 120. The Edit screen shows note starts and lengths in clicks, and the HIGH unit moves one click at a time.

5. Basic Concepts

Screens

RetroSeq has three screens, each a closer look at the song than the one before. The **Main** screen lists the tracks and their settings; **View** shows every bar of every track; **Edit** shows the notes of one bar. Windows such as OPTIONS open on top of whatever screen you are

on.



screen navigation · 3 screens, windows on top

Main is home: from View or Edit, [Esc] always brings you back.

Building a Song

A song is up to 64 tracks. Each track has its own output (a MIDI port or a VST3 instrument), MIDI channel and settings, and holds the notes and controller data you record. A typical way to work:

1. Set the tempo with **T**empo and the time signature in OPTIONS.
2. Give track 1 an output with [Ctrl]-O and a name with **N**ame.
3. Record it with **R**ecord, then [Spacebar].
4. Move to the next track and repeat, listening to what you already have.
5. Fix timing with the Qnt column, or single notes in the Edit screen.
6. Arrange sections in View with Copy and Insert.

Controlling the MIDI Network

One external synth can play several parts if each track uses its own MIDI channel on the same port. A VST3 instrument normally listens on every channel, so give each part its own VST3 track instead. The Prg, Vol and Pan columns set each instrument's sound, volume and position whenever playback starts, so a song always starts the same way.

Recording Techniques

- **From the start** (Main screen): **R** then [Spacebar] erases the track and records from bar 1, after the lead-in set in OPTIONS.

- **From any bar** (View screen): move the cursor to the bar, press [Ctrl]-R then [Spacebar]. Only the bars you play over are replaced.
- **Try again**: [Ctrl]-B (Backup) restores the track as it was before the take.
- **Tighten the timing** without changing what you played: set the Qnt column to 16 or 8. The original data is kept.

6. Tutorial

These three lessons take you from an empty song to a finished, arranged and corrected piece. Each takes about fifteen minutes. You need one keyboard and at least one sound source: an external synth, a General MIDI module or VST3 instruments. Where a step names a key, press that key; where it names a value, type it and press [Enter].

When a prompt shows a suggested answer (highlighted), the first key you type replaces it. Press [Enter] alone to accept it.

Lesson 1 — Record a First Song

You will record four bars of drums, loop them, add a bass line and then a melody from bar 5.

Prepare the song

1. Press **F**, choose *New* and confirm with **Y**. All tracks are cleared; outputs you set up earlier are kept.
2. Press **T**, type `100` and press [Enter]. The status area shows BPM 100.0.
3. Press **O** to open OPTIONS. Check that *Metronome* is *Rec* and *Lead-in* is *1*. Press [Esc].

Track 1: drums

4. The cursor is on track 1. Press **N**, type `Drums`, [Enter].
5. Press [Ctrl]-O and choose your drum sound: a VST3 drum instrument, or the MIDI port of a General MIDI module.
6. For a General MIDI module, move to the **Ch** column with → and type `10` [Enter]: drums are on channel 10.
7. Play a few notes on the keyboard. You hear the drums at once, because the track under the cursor always receives what you play.
8. Press **R**. REC flashes and the message line shows *Record track 1 from the start*.
9. Press [Spacebar]. The metronome counts one bar (the Bar field counts from 0:1), then recording starts at 1:1.
10. Play a four-bar beat and press [Spacebar] when the fifth bar begins.

11. Press [Spacebar] to listen. Not happy? Press **R** and [Spacebar] to record it again; the new take replaces the old one.

Loop the drums

12. Press **L**. The Loop column shows LOOP and the four bars now repeat for as long as the song plays. The loop ends after the bar of the last note that starts, so a cymbal ringing into bar 5 does not make it a five-bar loop.

Track 2: bass

13. Press ↓ to move to track 2, **N**, `Bass`, [Enter].
14. [Ctrl]-O, choose a bass sound. For a multi-timbral synth use the same port as the drums and set **Ch** to `2`.
15. **R**, [Spacebar]: the drums play while you record. Play eight bars of bass and press [Spacebar].
16. Missed the start? Press [Ctrl]-B. The track comes back as it was before the take (here: empty), and you can record again.

Track 3: melody from bar 5

17. Move to track 3, name it `Lead` and give it an output.
18. Press **V** for the View screen. Press → four times so the cursor is on bar 5 of track 3.
19. Press [Ctrl]-R, then [Spacebar]. The lead-in plays bar 4; recording starts at bar 5.
20. Play four bars and press [Spacebar]. The View grid shows ■ in bars 5–8 of Lead.

Save

21. Press **F**, choose *Save as*, type a name such as `Lesson1` and save. RetroSeq writes `Lesson1.mid` and `Lesson1.rsq`.

Lesson 2 — Arrange with Buffers

You will turn the eight bars from Lesson 1 into a 16-bar piece: repeat the bass, copy the melody and add a break.

Repeat the bass

1. In the View screen, move to track 2 (Bass), bar 1.
2. Press **C**. Move the cursor to bar 8: bars 1–8 are highlighted. Press [Enter].
3. At *Buffer (0-9 or T)* type `1` and press [Enter]. The message line confirms *Copied bars 1-8 to buffer 1*.
4. Move to bar 9 of the Bass track. Press **I**, type `1` [Enter], then [Enter] at *Number of times* (once). The bass now runs to bar 16.

Copy the melody

5. Move to bar 5 of track 3 (Lead). Press **C**, move to bar 8, [Enter], [Enter] (buffer T).
6. Move to bar 13 of Lead and press **R**, [Enter]. The melody is copied over the empty bars 13–16 without moving anything after it.

Add a break

7. Move to bar 9 of the Drums track and press **L** to switch looping off, so the drums only play their four bars.
8. To make room for a one-bar break in the bass, move to bar 9 of Bass and press [Ins]. An empty bar appears and everything from bar 9 moves one bar later.
9. Changed your mind? [Ctrl]-B restores the Bass track as it was before [Ins].

Check the result

10. Press [Ctrl]-[Home] and [Spacebar]. In View, the cursor follows the music bar by bar.
11. Save with **F**, *Save*.

Lesson 3 — Correct Notes in the Edit Screen

You will fix a wrong note, lengthen one, delete one and add one in the melody.

1. In the View screen, move to a bar of the Lead track that needs work and press **E**. The Edit screen shows that bar.
2. Press **U** until *Units* shows 16. The cursor now moves in sixteenth notes.
3. Press [Tab] to jump from note to note. The note under the cursor turns yellow and its pitch, velocity, start and length appear top right.

A wrong pitch

4. On the wrong note, press **P**. *PITCH* flashes.
5. Press **+** or **−** until it is right; [ ] moves it by an octave. Press → to keep the change.

A note that is too short

6. On the note, press **L**, then **+** once for each sixteenth it should be longer. Press → to keep it.

A note to remove or add

7. On an extra note, press [Del].
8. To add a note with the same length and velocity as an existing one, put the cursor on that note and press *****. Move to the empty spot (↑↓ for pitch, ←→ for time) and press [Ins].
9. For a note of a fixed value, use [Ctrl]-Q (quarter), [Ctrl]-E (eighth) or [Ctrl]-S (sixteenth) instead.

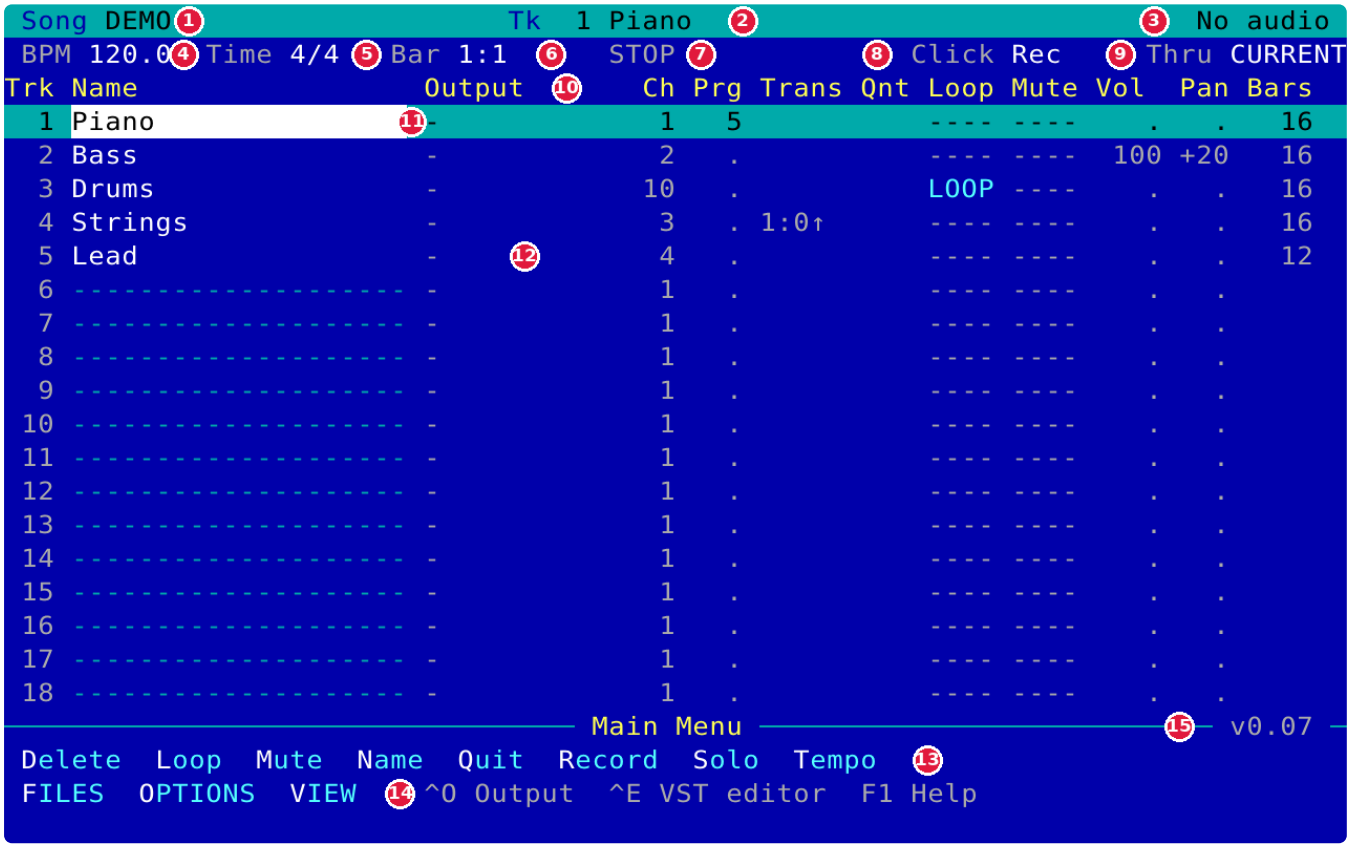
Listen and go back

10. Press [Spacebar] to play from the start of the bar, and again to stop.
11. [Home] and [End] move to the previous and next bar. [Esc] returns to the View screen.

Any change you regret can be undone with [Ctrl]-B, which also works on the Main and View screens.

7. Main Screen

The Main screen is where RetroSeq starts and where every track's settings live. Press [Esc] repeatedly from any screen to come back here.



1. **Song** — name of the current song file
2. **Tk** — current track number and name
3. **Audio device** — sample rate and buffer size, or *No audio*
4. **BPM** — tempo
5. **Time** — time signature
6. **Bar** — position as bar:beat; counts from 0 during a lead-in
7. **Transport** — STOP, PLAY or REC (REC flashes while armed)
8. **Click** — metronome mode
9. **Thru** — MIDI Thru CURRENT or OFF
10. **Column headings** of the work area
11. **Field cursor** — the field that + − [ ] . and digits change

12. **Unused track** — dashes instead of a name
13. **Menu commands** — press the first letter
14. **Screens and windows**, and the main Ctrl keys
15. **Version number**

Status Area

Field	Meaning
Song	Name of the current song file (NEW-SONG until saved)
Tk	Current track: number and name
BPM	Tempo, set with **T**empo
Time	Time signature, set in OPTIONS
Bar	Song position as bar:beat. During a lead-in it counts up from 0
STOP / PLAY / REC	Transport state. REC flashes when Record is armed and waits for [Spacebar]
SOLO n	Flashes when a track is soloed
Click	Metronome: Off, Rec (only while recording) or Always
Thru	MIDI Thru: CURRENT (the track under the cursor plays what you play) or OFF
Hz / samples	Audio device sample rate and buffer size

Work Area

Move between tracks with ↑↓, [PgUp]/[PgDn] and [Tab]. Move between columns with ←→. On the highlighted field, **+ −** change the value by one, **[ ]** by ten, digits + [Enter] type a value, and **.** removes a Prg, Vol or Pan setting.

Column	What it does	How to change it
Trk	Track number, 1–64	—
Name	Track name, up to 20 characters. Empty tracks show dashes	**N**ame or [Enter]

Column	What it does	How to change it
Output	`M:` MIDI port or `V:` VST3 instrument	[Ctrl]-O, [Enter] or + −
Ch	MIDI channel 1–16 that all events are sent on	+ − [ ], digits, [Ctrl]-C
Prg	Program sent at the start of playback; `.` = none. On a VST3 track it selects the plug-in's program	+ − [ ], digits, [Ctrl]-P
Trans	Transpose in octaves:semitones with an arrow, e.g. 1:0↑. Does not change the data	+ − (semitone), [ ] (octave)
Qnt	Playback quantize 4, 4t … 64, 64t. Does not change the data	+ −
Loop	LOOP repeats the track in whole bars	**L**oop
Mute	MUTE silences the track; SOLO plays only this track	**M**ute, **S**olo
Vol	MIDI volume (CC7) 0–127, or velocity scaling −127…+127	+ − [ ], [Enter] toggles the mode
Pan	Pan (CC10) −64…+63, 0 = centre	+ − [ ], digits
Bars	Length of the track in bars	—

Vol and Pan on MIDI tracks are sent as MIDI volume (CC7) and pan (CC10) at the start of playback and whenever you change them; the synth decides whether it responds. Instruments keep the last value they received, so set them back explicitly rather than with `.` if a synth stays too quiet.

Vol and Pan on VST3 tracks (version 0.09 and later) are applied by RetroSeq itself to the instrument's audio output, because VST3 has no standard way to receive MIDI volume or pan and many plug-ins ignore CC7 and CC10. Vol 127 is full level, 64 about a quarter (the MIDI volume curve), 0 silent; Pan balances the instrument's left and right outputs. Velocity scaling works the same on both kinds of track.

Column Examples

Trans, Qnt and Loop change what you hear, not what is stored. Switch them off and the track plays exactly as recorded; save the song and the recorded notes are saved unchanged, with the column values in the `.rsq` file.

Trans — play a part in another key

The demo's Strings play C5 and G5 in bar 5. Put the cursor on the Trans column of track 4 and press **]** once.

	Trans column	Stored notes	Heard
Before	(empty)	C5, G5	C5, G5
After]	1:0↑	C5, G5	C6, G6
After] and – – – – –	0:7↑	C5, G5	G5, D6

What you play on the keyboard while that track is selected is transposed too, so it matches what you hear.

Qnt — tighten timing without losing the feel

A hi-hat recorded slightly late: notes at 1:1 +18 clicks, 1:2 −9 clicks, 1:3 +25 clicks (480 clicks per quarter note).

Qnt	Heard at	Stored at
(empty)	+18, −9, +25	+18, −9, +25
16	exactly on 1:1, 1:2, 1:3	+18, −9, +25
8t	nearest eighth-note triplet	+18, −9, +25

Note lengths are kept: a note moved earlier also ends earlier. Set Qnt back to empty with **.** to hear the original playing again.

Loop — repeat a pattern

The demo's Drums are 16 bars long. With LOOP they repeat from bar 17 for as long as the song plays: bar 17 sounds like bar 1, bar 33 like bar 1 again. A one-bar drum pattern with LOOP fills a whole song. In the View screen an **L** marks the end of the looping part.

Vol — volume or velocity?

Vol shows	Mode	What is sent	Use it when
`.`	none	nothing	the instrument's own volume is right
`100`	MIDI volume	CC7 = 100 at start and on change	the synth responds to CC7; all tracks on that channel change
`+20`	velocity scaling	every note's velocity + 20 (1–127)	the synth ignores CC7, or two tracks share a channel

On a VST3 track, `100` sets the instrument's output level directly instead of sending CC7, so it works with every plug-in. Press [Enter] on the Vol column to switch between the two modes. Velocity scaling also changes the tone on velocity-sensitive sounds: a piano at −40 sounds softer, not just quieter.

Prg and Pan

Prg 5 sends program change 5 when playback starts. On a VST3 track it selects the plug-in's program 5 instead, if the plug-in has a program list ([Ctrl]-P shows the names). Pan −64 is hard left, 0 centre, +63 hard right.

Menu Commands

Command	Function
Delete	Erase the current track (Y) or all tracks (A). [Ctrl]-B restores a single track
Loop	Toggle looping for the current track
Mute	Toggle mute; works during playback
Name	Type a track name
Quit	Exit RetroSeq; press Q again to confirm
Record	Arm recording on the current track
Solo	Play only the current track; S on another track moves the solo
Tempo	Type a tempo, 16–400 BPM; works during playback
Punch-in	Record only inside a bar range on the current track (see chapter 8)
XFORMS	Transform the whole current track (chapter 10)
E	EDIT screen at the current bar
FILES	Load, save, save as, new song
OPTIONS	Options window
VIEW	View screen

Record and Playback from the Main Screen

Playback from the Main screen always starts at bar 1; use the View screen to start elsewhere.

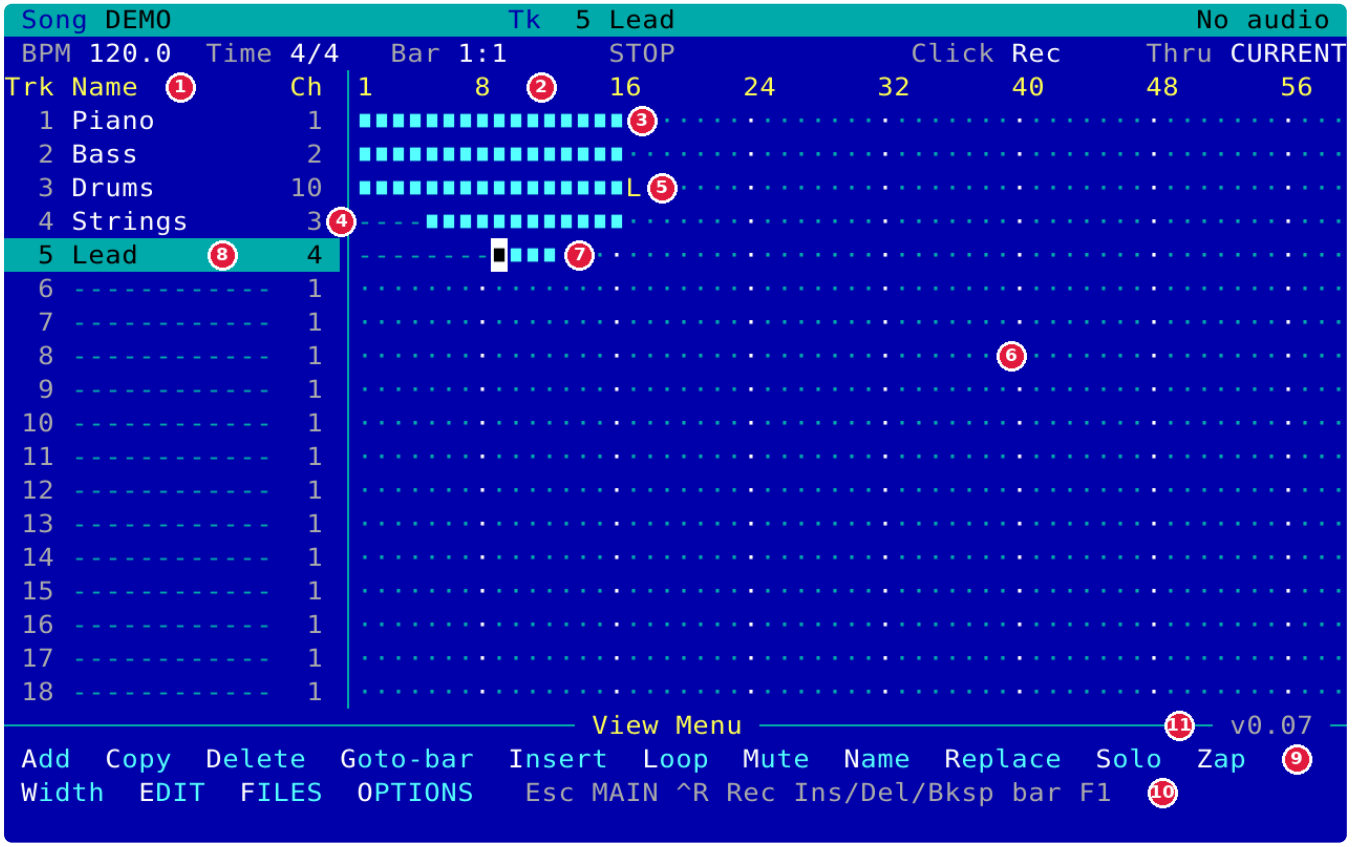
To record a track

1. Move the cursor to the track and give it an output with [Ctrl]-O.
2. Press **R**. REC flashes and the message line confirms the track.
3. Press [Spacebar]. After the lead-in (one bar by default) recording starts at bar 1.
4. Press [Spacebar] to stop. The take replaces everything that was on the track.

Example: record the Lead part over the demo song. Move to track 5, press **R**, then [Spacebar], play along from bar 1, and press [Spacebar] when done. Not happy? [Ctrl]-B brings the old Lead back.

8. View Screen

The View screen shows the whole song as a grid of bars, one row per track. Use it to see the arrangement, to play or record from any bar, and to move bars around with the buffers. Press **V** on the Main screen to enter and [Esc] to go back.



1. **Track list** — number, name and channel (hidden with **W**idth)
2. **Bar numbers** every eighth bar; those columns are brighter
3. ■ a bar with notes

4. **–** a bar inside the track without notes
5. **L** — the looping part ends here
6. **·** after the end of the track
7. **Cursor** — track and bar
8. **Current track**
9. **View menu commands**
10. **Screens and keys**
11. **Version number**

Work Area

Every eighth bar is numbered along the top and its column is brighter. The status area is the same as on the Main screen.

Character	Meaning
■	Bar with notes
–	Bar inside the track but without notes
·	Bar after the end of the track
M	Track is muted (shown at bar 1)
L	Track loops; shown just after its last bar

Width hides the track names and channels so the grid fills the screen.

Moving the Cursor

Key	Moves
←→	One bar
[Ctrl] ←→	Eight bars
↑↓	One track
[Home] / [End]	One screen left / right
[Ctrl] [Home] / [End]	Start / end of the track
[PgUp] / [PgDn]	One screen of tracks up / down
[Ctrl] [PgUp] / [PgDn]	First / last track

Key	Moves
[Tab] / [Shift]-[Tab]	Next / previous track with data

Special Keys

Key	Function
[Ins]	Insert an empty bar at the cursor; later bars move right
[Del]	Remove the bar at the cursor; later bars move left
[Backspace]	Remove the bar to the left of the cursor

Menu Commands

Command	Function
Add	Insert a number of empty bars at the cursor; past the end of the track it fills up to the cursor with empty bars
Copy	Copy a range of bars into a buffer; the track is unchanged
Delete	Move a range of bars into a buffer; later bars close the gap
Goto-bar	Type a bar number to move the cursor there
Insert	Insert a buffer at the cursor, one or more times; later bars move right
Loop, **M**ute, **N**ame, **S**olo	As on the Main screen
Replace	Overwrite bars from the cursor with a buffer
Tempo	Type a tempo
Zap	Copy a range into a buffer and leave empty bars behind
Width	Toggle the full-width grid
BLOCK	Block Moves menu: copy, delete, insert, replace or zap a rectangle of tracks and bars

Command	Function
Punch-in	Arm punch-in recording for a bar range
XFORMS	Mark a bar range and transform it (chapter 10)
EDIT	Edit screen at the cursor bar

Using the Buffers

There are eleven buffers: **0–9** and **T** (Temp). A buffer holds bars of one track; inserting it puts the notes on the current track and channel, so you can copy a part from one track to another.

To copy bars into a buffer

1. Put the cursor on the first bar of the range and press **C** (or **D**, **Z**).
2. Move the cursor to the last bar. The range is highlighted.
3. Press [Enter], type the buffer (0–9 or T) and press [Enter].

To insert a buffer

1. Move the cursor to the bar where the bars should go.
2. Press **I**, type the buffer and press [Enter].
3. Type how many times to insert it and press [Enter].

Example: the demo's Lead plays in bars 9–12. To repeat it in bars 13–16, put the cursor on Lead bar 9, press **C**, move to bar 12, [Enter], [Enter] (buffer T). Then move to bar 13, press **R**, [Enter]: the four bars are copied over the empty bars 13–16.

Track Length and Empty Bars

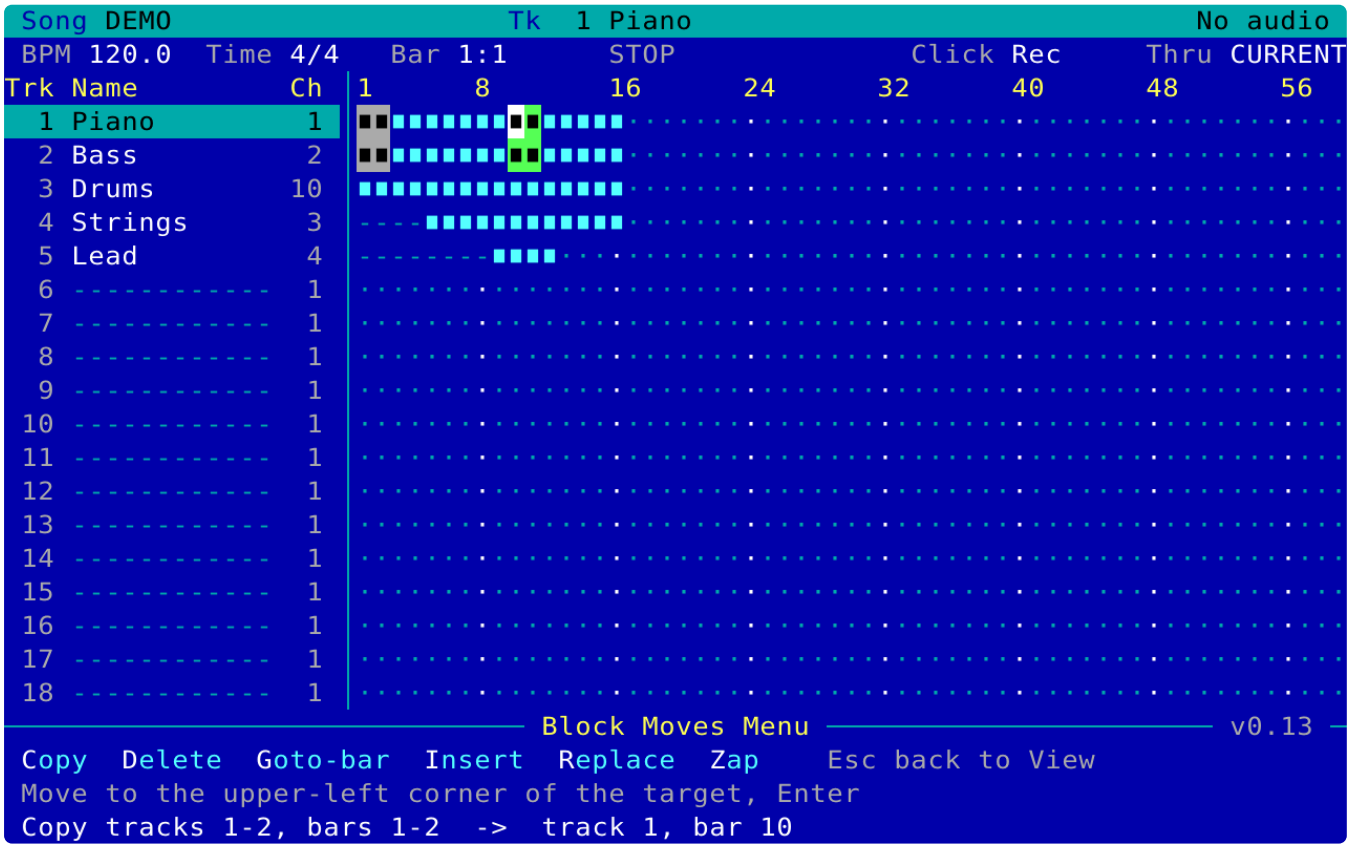
A track can end in empty bars. **[Ins]** or **A**dd with the cursor past the last bar fills the track with empty bars up to the cursor, then inserts the bars you asked for. Empty bars show as **–** in the grid.

The length is kept: Loop repeats the track over its full length, the `.rsq` file stores it, and the MIDI file gets its End of Track at that bar. **[Del]**, **[Backspace]** and Delete shorten it again.

Example: in the demo the Lead track has notes in bars 9–12, so it loops every 12 bars while the other tracks are 16. Put the cursor on bar 13 of Lead, press **A**, type `4`, [Enter]. Lead is now 16 bars; with **L**oop on it stays in step with the rest.

Block Moves Menu

Press **B** on the View screen; [Esc] returns to the View menu. Block Moves work on a rectangle of tracks *and* bars, where the buffers work on one track. Nothing goes into a buffer, and every command asks for confirmation before it changes the song.



Command	Function
Copy	Insert the source block at the target one or more times; later bars move right
Delete	Remove the bars from the tracks in the range; later bars move left
Goto-bar	Type a bar number to move the cursor there
Insert	Insert as many empty bars as the range is wide, on the tracks in the range
Replace	Overwrite the target with the source block
Zap	Erase the range and leave empty bars

To copy a block

1. Press **B**, move the cursor to one corner of the source and press **C**.

2. Move to the opposite corner (←→ bars, ↑↓ tracks) and press [Enter]. **R** resets the range to the cursor; **K** keeps the previous range.
3. Move to the upper-left corner of the target, shown in green, and press [Enter].
4. Type how many times to insert the block, [Enter], then **Y** or [Enter] to confirm.

Example: to repeat the first two bars of Piano and Bass at bar 10, press **B**, put the cursor on Piano bar 1, **C**, move to Bass bar 2, [Enter], move to Piano bar 10, [Enter], `2`, [Enter], **Y**. Both tracks grow by four bars. [Ctrl]-B undoes it on both tracks.

Record and Playback from the View Screen

Unlike the Main screen, View plays and records from the cursor. During playback the cursor follows the music bar by bar.

To record from any bar

1. Move the cursor to the bar where recording should start.
2. Press [Ctrl]-R. REC flashes.
3. Press [Spacebar]; the lead-in plays the bar(s) before the cursor, then recording starts.
4. Press [Spacebar] to stop. Only the bars you played over are replaced; the rest of the track is kept.

Punch-In

Punch-in replaces a few bars in the middle of a take without touching the rest. Press **P** on the Main or View screen and type the punch bars, for example `9-12`. The status line blinks PUNCH.

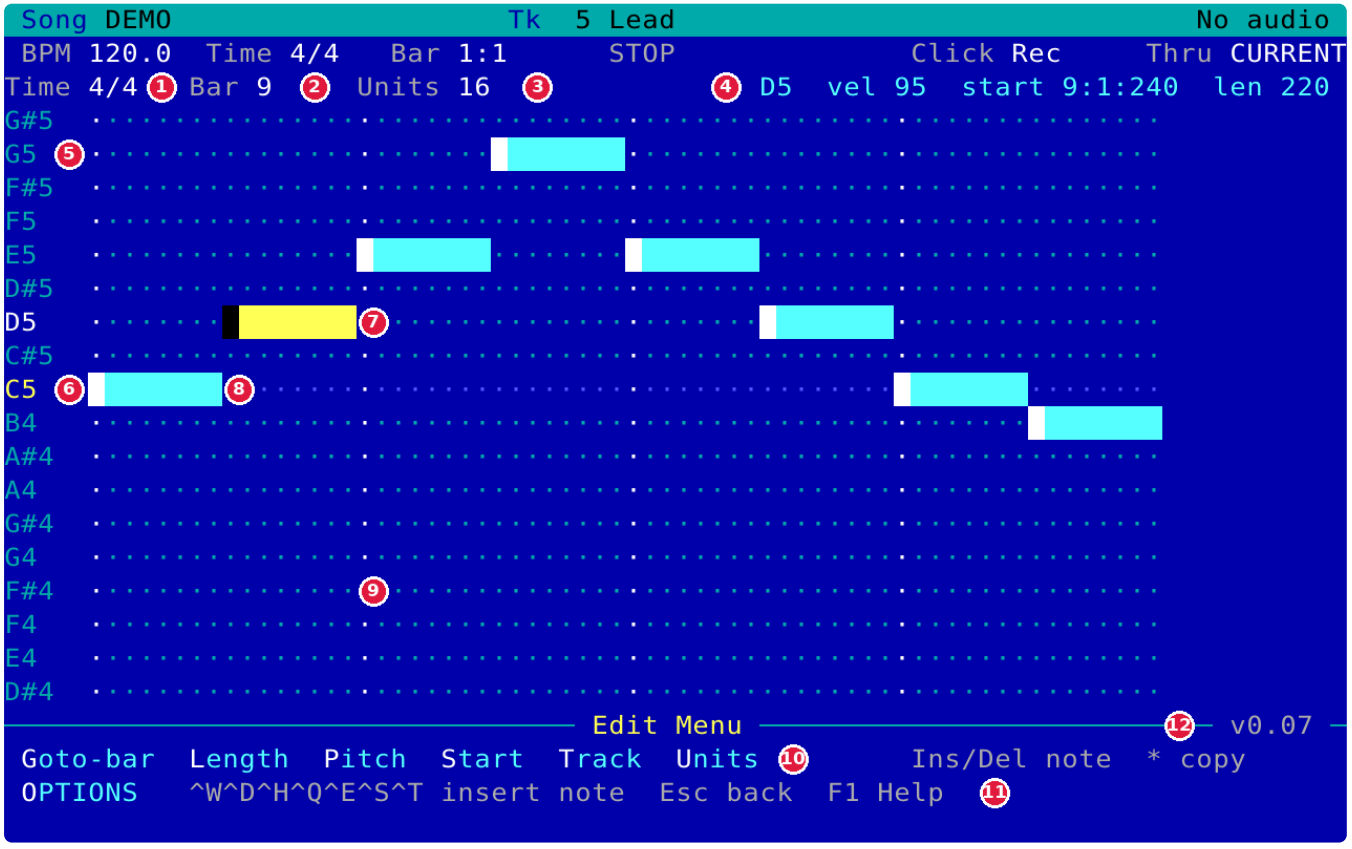
1. Press [Space]. Playback starts the lead-in number of bars before the punch-in point (at least one bar), so you can play along.
2. Play. Only notes played inside the punch bars are recorded; the track's old notes in those bars are silent while you play.
3. Press [Space] to stop. The punch bars now hold the new take; everything before and after is unchanged.

Notes still held at the punch-out point are closed there. [Ctrl]-B restores the track as it was before.

Example: the bass in bar 10 of the demo has a wrong note. Put the cursor on Bass, press **P**, type `10`, [Enter], then [Space]. Bar 9 plays as a lead-in, you play bar 10, and RetroSeq goes on playing bar 11 while recording stops.

9. Edit Screen

The Edit screen shows the notes of one bar of the current track, so you can move, lengthen, transpose, add and delete single notes. Press **E** on the Main or View screen to enter at the current bar, and [Esc] to go back.



1. **Time** — time signature of the bar
2. **Bar** — bar number
3. **Units** — how far ←→ move the cursor
4. **Note under the cursor** — pitch, velocity, start, length
5. **Pitch names**, one row per semitone
6. **C rows** are yellow; the cursor row is white
7. **Note under the cursor**, in yellow
8. **A note** — white start column, then its length
9. **Beat columns** are brighter
10. **Edit menu commands**
11. **Keys for inserting notes**
12. **Version number**

Work Area

- **Vertically: pitch.** Each row is one semitone, names on the left (C4 = middle C, MIDI note 60). C rows are marked in yellow. Arrows ↑ ↓ in the name column mean there are notes above or below what is shown.
- **Horizontally: time.** The bar is divided into 64th notes, one column each (coarser if a long bar would not fit). The brighter dot columns are the beats.
- **Notes** are solid bars from start to end; the first column is white. The note under the cursor is yellow and its pitch, velocity, start and length are shown in the top right corner.

In the 50- and 60-line screen modes more pitches fit on screen.

Time Units

The cursor moves in time by the current unit, shown as *Units* at the top. **U** steps through them; the insert-note length follows the unit.

Unit	Normal	Triplet
Quarter note	4	4T
Eighth note	8	8T
Sixteenth note	16	16T
32nd note	32	32T
64th note	64	64T
Single click (1/480 quarter note)	HIGH	—

Moving the Cursor

Key	Moves
↑↓	One semitone
←→	One time unit; past the bar's edge into the next or previous bar
[Ctrl] ←→	Four time units
[Home] / [End]	Previous / next bar
[Ctrl] [Home] / [End]	First / last bar of the track

Key	Moves
[PgUp] / [PgDn]	One octave up / down
[Ctrl] [PgUp] / [PgDn]	Nearest note above / below in this bar
[Tab] / [Shift]-[Tab]	Next / previous note

Menu Commands

Command	Function
Goto-bar	Type a bar number
Length	Change the length of the note under the cursor
Pitch	Move the note under the cursor up or down
Start	Move the note under the cursor earlier or later
Track	Type a track number; stays on the same bar
Units	Next time unit
KBD_ENTRY	Step Record from the MIDI keyboard; K or [Esc] ends it
MIDI edit	Every event on the track as an editable list
OPTIONS	Options window

Manipulating Notes

To change a note

1. Put the cursor on the note; it turns yellow.
2. Press **L**, **P** or **S**. The mode name flashes at the top.
3. Use the keys below. The note changes as you press them and you hear it on the next playback.
4. Any other key keeps the change. [Esc] while the mode is flashing puts the note back.

Keys	Length and Start	Pitch
+ −	One time unit	One semitone
[ ]	Four time units	One octave

Keys	Length and Start	Pitch
< >	One click	One semitone

To insert notes

- [Ins] inserts a note at the cursor with the insert length and velocity, shown top right when the cursor is on empty space.
- ***** copies the length and velocity of the note under the cursor into the insert values.
- Notes of a fixed value go in with one key:

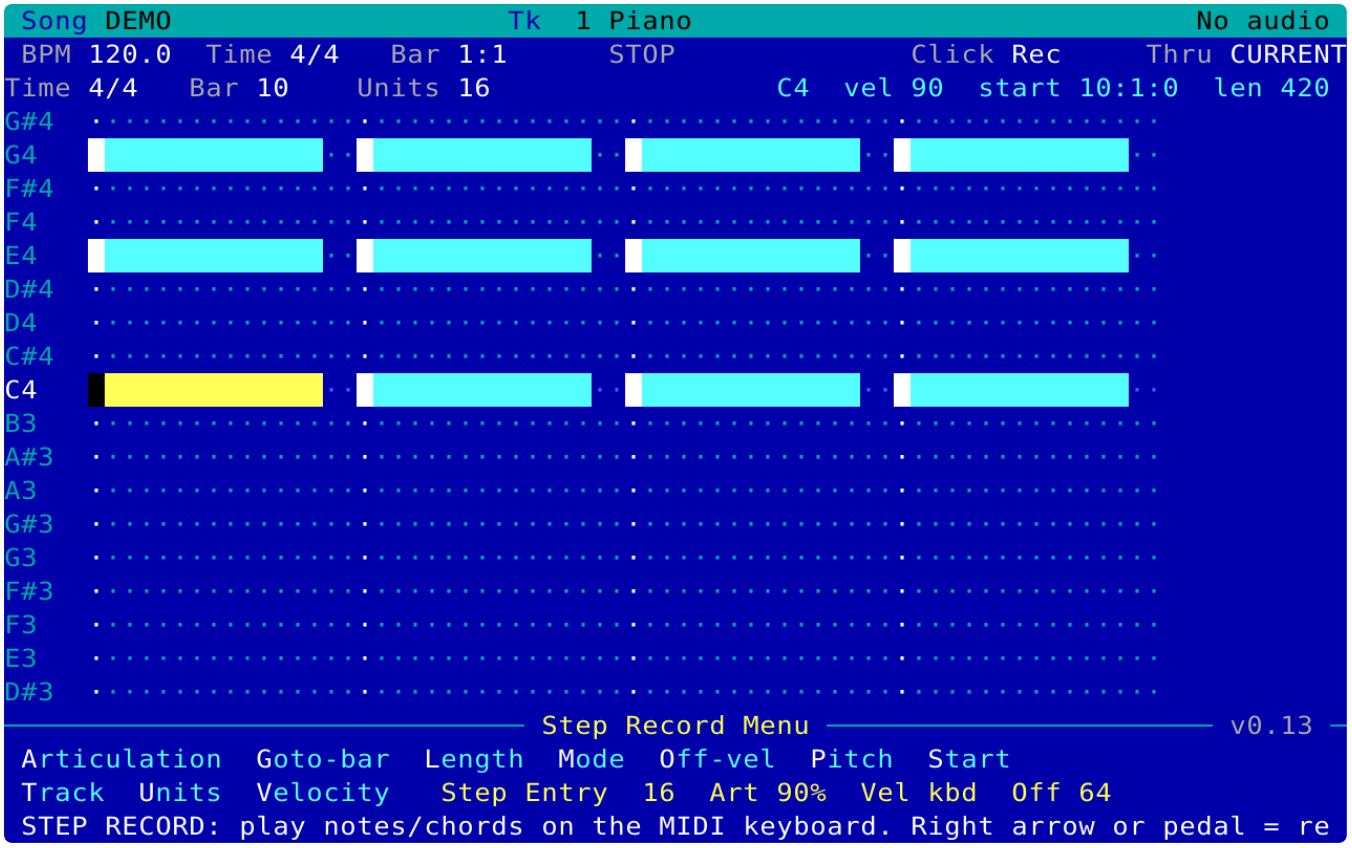
Keys	Note
[Ctrl]-W	Whole
[Ctrl]-D	Dotted half
[Ctrl]-H	Half
[Ctrl]-Q	Quarter
[Ctrl]-E	Eighth
[Ctrl]-S	Sixteenth
[Ctrl]-T	32nd

To delete a note: put the cursor on it and press [Del].

Example: in bar 9 of the demo's Lead, the D5 on the second eighth is a little short. Put the cursor on it with [Tab], press **L**, then **+** once (units 16): it now lasts a sixteenth longer. Press → to keep it.

Step Record

Step Record enters notes from the MIDI keyboard at a fixed rhythm, however you play them. Press **K** (KBD_ENTRY) on the Edit screen; **K** or [Esc] ends it. The menu changes to the Step Record menu and its last line shows the settings.



- **A note** is entered at the cursor with the length *Units × Articulation %*. When you release it the cursor moves one unit.
- **A chord**: hold the first key and press the others; all go in at the same position. Re-strike a held key to take that note out.
- **A rest**: the right arrow or the sustain pedal moves one unit without a note.

Command	Function
Articulation	Note length as 1–100 % of the step; 50 % with 8th-note units gives 16th notes
Mode	Step Entry → Edit Existing → Preserve Rhythm
Off-vel	Note-off velocity from the keyboard, or a fixed value
Velocity	Note-on velocity from the keyboard, or a fixed value
Units	Step length, 4 to 64T
Goto-bar, **L**ength, **P**itch, **S**tart, **T**rack	As on the Edit screen

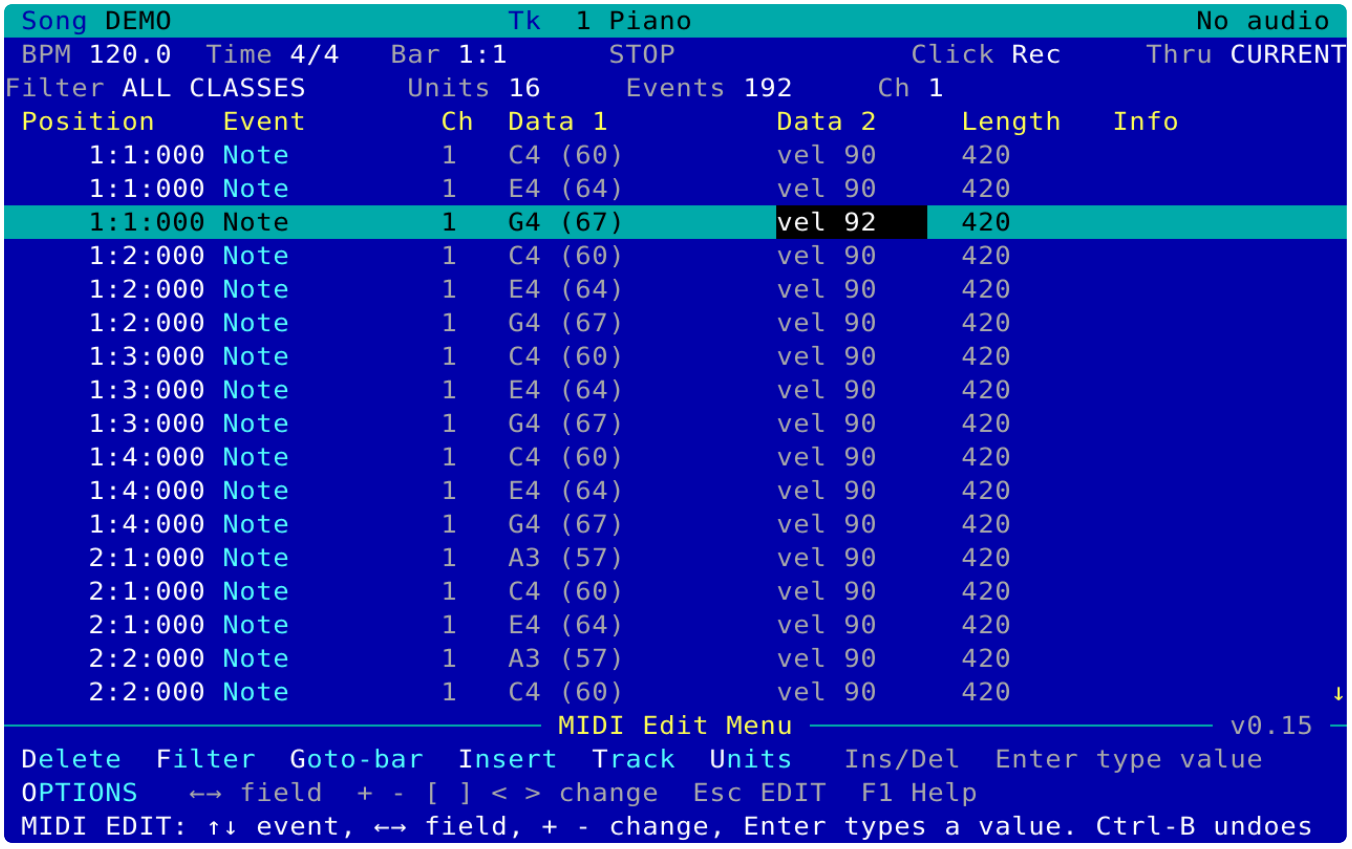
Edit Existing: a key changes the pitch and velocity of the note under the cursor; [Tab] or the pedal goes to the next note. **Preserve Rhythm** does the same and then moves to the next note by itself, so you can play a new melody on an existing rhythm.

MIDI input is read in the audio callback, so the audio device must be running. [Ctrl]-B restores the track as it was before Step Record started.

Example: a bass line in 8th notes. Set **U**nits to 8, press **K**, then play C2, C2, rest (right arrow), G1 … Each key fills one 8th; with Articulation 90 % the notes leave a short gap.

MIDI Edit Screen

The MIDI Edit screen lists every event on the track: notes with their length, controllers, program changes, pitch bend, channel and poly pressure, and sysex. Press **M** on the Edit screen; [Esc] returns to the Edit screen at the bar of the selected event. Note-offs are not listed; they follow their note through the Length field.



Column	Note	Controller	Program	Pitch bend	Pressure
Position	bar:beat:tick	bar:beat:tick	bar:beat:tick	bar:beat:tick	bar:beat:tick
Data 1	pitch	controller number	program 0–127	−8192 to +8191	value (poly: note)
Data 2	velocity	value	—	—	poly: value
Length	ticks	—	—	—	—

Keys

Key	Function
↑ ↓, [PgUp] [PgDn], [Home] [End]	Select an event
← →	Select a field
+ −	Change by one; position and length by the current unit, pitch bend by 128
[ ]	Change by ten, an octave for pitch, four units for time
< >	Change position or length by one click
[Enter]	Type a value: `9:2:240`, a note name, a number, or a length such as `16` or `120c`
[Ins] or **I**	Insert a note, controller, program change, pitch bend or pressure event at the selected position
[Del] or **D**	Delete the event (a note's note-off goes with it)
Filter	Show one class: notes, controllers, one controller number, program, pitch bend, aftertouch, sysex
Goto-bar, **T**rack, **U**nits	Jump to a bar, change track, change the time unit

[Ctrl]-B restores the track as it was when you opened MIDI Edit, or before the last insert or delete.

Example: add a mod-wheel swell to the Lead in bar 9. Press **M**, **F**ilter → *One controller number* → `1`. Press [Ins], choose *Controller*, then → to Data 2, [Enter] and type `0`. Insert another two beats later with value `100`.

10. XFORMS (Transforms)

Transforms change the recorded data of a track in one go: timing, velocity, pitch and length. Unlike the Trans and Qnt columns they are destructive, so [Ctrl]-B is the way back.

Using Transforms

- **X** on the Main screen transforms the whole current track.
- **X** on the View screen marks a bar range like the buffer commands: move the cursor to the end of the range and press [Enter].

Then pick a transform from the list and answer its questions. Each answer is remembered for the rest of the session, so repeating a transform is a matter of [Enter].

Times are typed as note values: `16` = a 16th, `8T` = an 8th triplet, `4.` = a dotted quarter, `120c` = 120 clicks (480 clicks per quarter note), `-8` = an 8th earlier. **Pitches** are names or numbers with C4 = 60: `C1`, `D#1`, `Bb3`, `36`; a range is `36-38`, `C1-D1` or `all`.

Time Transforms

Transform	Asks for	What it does
Shift time	amount	Moves notes earlier or later; controllers and pitch bend in the range move too
Scale time	%	Stretches or compresses from the start of the range; 200 % = twice as long
Super-Quantize	grid, strength %, swing %, lengths Y/N	Pulls notes towards the grid. 100 % = exactly on the grid, 50 % = half way. Swing 50 % is straight, 66 % a triplet feel, 75 % the maximum. Lengths Y also rounds note lengths to the grid
Humanize	max time, max velocity	Moves each note a random amount within the limits

Velocity Transforms

Transform	Asks for	What it does
Velocity scale	%	Multiplies every velocity; 80 % softer, 120 % harder
Velocity add	−127 to 127	Adds a fixed amount
Velocity set	1–127	Gives every note the same velocity
Velocity compress	%	Squeezes the dynamics around the average; 50 % halves the spread, 150 % widens it
Velocity ramp	start–end	A crescendo or diminuendo across the range, e.g. `40-120`

Velocities are always kept within 1–127.

Pitch and Length Transforms

Transform	Asks for	What it does
Transpose	semitones, pitch range	Moves notes up or down; the range limits it to some notes, e.g. only `C3-B4`
Invert	centre note	Mirrors the melody around a note
Length set	length	Gives every note the same length
Length scale	%	50 % for staccato, 150 % for longer notes
Legato	gap	Extends each note to the next note, minus the gap

Split Transforms

Transform	Asks for	What it does
Extract notes	pitch range, track	Moves the notes to another track, by default the next empty one
Delete notes	pitch range	Removes the notes

These are made for drum tracks: with General MIDI drums `36` is the kick, `38` the snare and `42-46` the hi-hats. [Ctrl]-B after Extract restores both tracks.

Examples

Swing a hi-hat part: on the View screen, mark the drum bars with **X**, choose *Super-Quantize*, grid `16`, strength `100`, swing `60`, lengths `N`.

Give the kick its own track: on the Main screen put the cursor on Drums, press **X**, choose *Extract notes*, type `36`, [Enter] and accept the suggested empty track.

A crescendo into the chorus: mark bars 13–16 of Strings with **X** on the View screen and choose *Velocity ramp* `40-110`.

11. Global Functions

Files

Press **F** (FILES) on the Main or View screen.

Item	Function
Load	Open a standard MIDI file (.mid, type 0 or 1)
Save	Save under the current name
Save as	Save under a new name
New	Clear all tracks and set 120 BPM, 4/4
Export	Save the current track as a MIDI file (type 0)
Import	Load one part of a MIDI file into the current track

Keeping your instruments (version 0.10 and later). When any track has an output, *Load* and *New* ask what to do with the instruments:

Command	Y	N
Load	Keep the current outputs and VST3 instruments; only the notes, names, channels and column values are loaded	Use the song's own instruments from its `.rsq` file; without one, the tracks get no output
New	Keep outputs, instruments and track names, ready for the next song	Clear outputs, instruments and track names too

[Esc] or any other key cancels. Keeping instruments is handy when you work through several songs with the same studio setup.

When loading, a type 0 file is split into one track per MIDI channel, track names and the first tempo and time signature are read, and the file's resolution is converted. Saving writes a type 1 file with one track per RetroSeq track.

Next to every saved `.mid`, RetroSeq writes a `.rsq` file with what a MIDI file cannot hold: each track's output, VST3 instrument and its sound settings, and the Prg, Trans, Qnt, Loop, Vol and Pan columns. It is read automatically when the song is loaded. Keep the two files together.

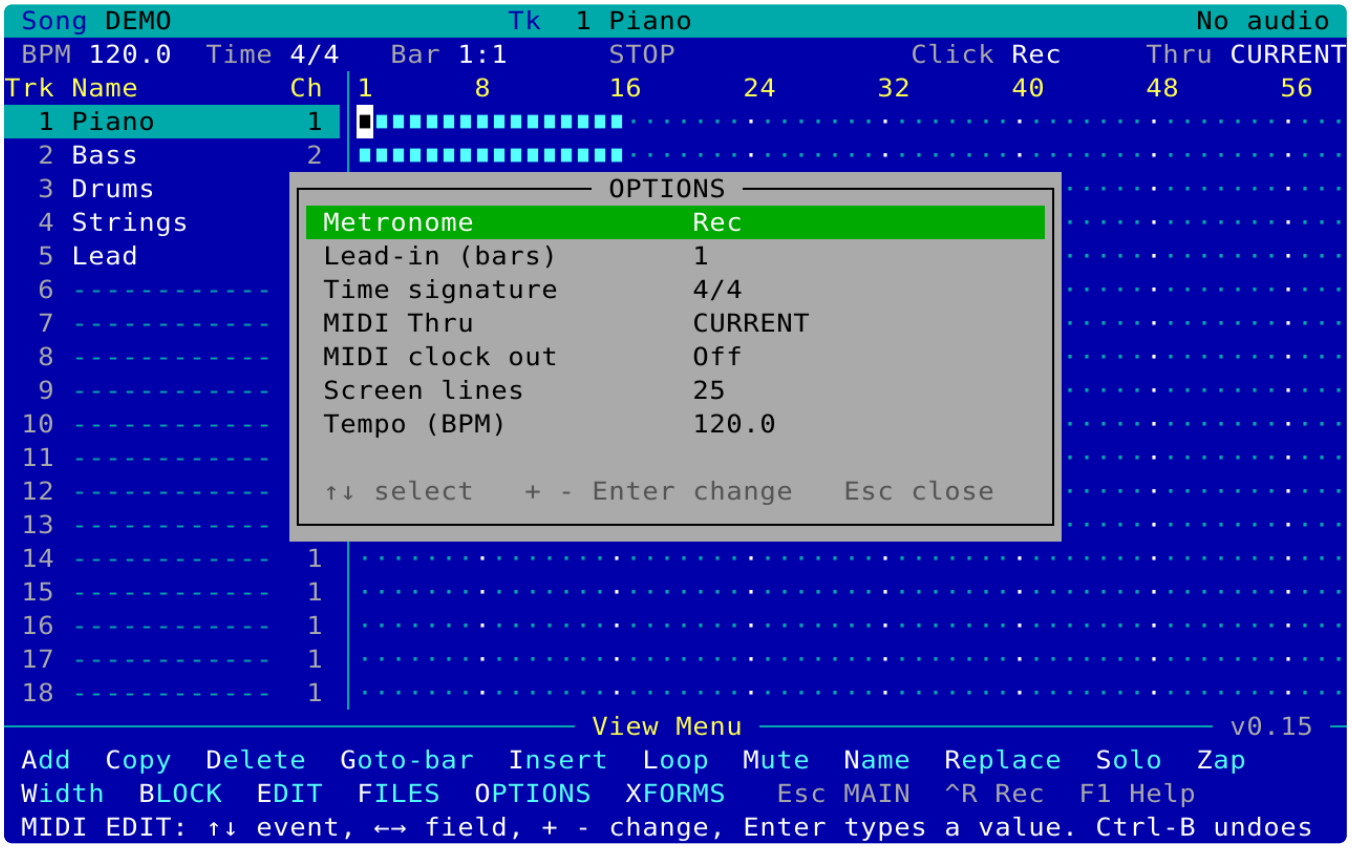
Export and import a single track (version 0.12). Two more choices in the FILES list move one track in or out, which is handy for drum patterns.

- **Export** saves the current track as a type 0 MIDI file with its name, tempo, time signature and length (End of Track), so trailing empty bars survive. If Trans, Qnt, Vol, Prg or Pan are set, **Y** saves the track *as played* with them applied, **N** the raw data.

- **Import** loads one part into the current track. The file is split per MIDI channel; if it holds more than one, pick a part from the list (name, channel, number of notes). The notes go to the track's own channel and output, MIDI or VST3. On a track that already has data, **R** replaces and **M** merges, e.g. a fill on top of a beat. The file's tempo is ignored, and [Ctrl]-B undoes the import.

Options Window

Press **O**, **[F3]** or **[Shift]-[F2]**. Move with ↑↓, change with + − or [Enter], close with [Esc].



Option	Values	Function
Metronome	Off, Rec, Always	Click on every beat, accented on the downbeat. Rec = only while recording, including the lead-in
Lead-in (bars)	0–4	Bars counted in before recording starts
Time signature	e.g. 4/4, 3/4, 7/8	Sets bar length for the whole song; [Enter] to type it
MIDI Thru	CURRENT, OFF	CURRENT plays what you play on the track under the cursor
MIDI clock out	Off, or ticked MIDI outputs	Sends clock, Start/Stop/Continue and Song Position Pointer; see MIDI Clock Out

Option	Values	Function
Screen lines	25, 43, 50, 60	Text lines on screen; more lines show more tracks and pitches
Tempo (BPM)	16–400	Same as **T**empo

The classic sequencers offered 43- and 50-line modes on EGA and VGA screens and up to 60 lines on some VGA cards. In RetroSeq these modes simply divide the window into more lines; make the window taller to keep the text large.

Configuration

Press **[F4]** for the two setup dialogs: *Audio and MIDI devices* and *VST3 plug-ins (scan)*. See chapter 3.

Metronome

[F2] switches the metronome off and back on to its last mode. The click is a short tone on the audio output, higher on the first beat of each bar. It follows the time signature.

MIDI Thru

With MIDI Thru CURRENT, everything you play is sent to the output of the track under the cursor, on that track's channel, with its Trans and Vol settings. While recording, the track being recorded receives MIDI in, wherever the cursor is. **[Shift]-[F9]** switches Thru off, for example when your keyboard has its own sounds switched on.

Play Range

[F5] plays part of the song. Type the bars, e.g. `9-12`, then **Y** to loop or **N** to play once; the status line shows RANGE. [Space] stops. On the View screen the suggestion starts at the cursor bar, which makes it quick to rehearse a section or audition a loop while you edit.

MIDI Clock Out

RetroSeq can drive drum machines, arpeggiators and other sequencers with MIDI clock, the job the sync jack on the V-24s did. In the Options window choose *MIDI clock out* and tick one or more MIDI outputs with [Enter]; *Done* closes the list. The choice is saved with the settings.

When	RetroSeq sends
Playback starts at bar 1	Start, then 24 clocks per quarter note

When	RetroSeq sends
Playback starts elsewhere	Song Position Pointer, Continue, clocks
The position jumps, or a Play Range loop restarts	Stop, Song Position Pointer, Continue
Playback stops	Stop

The clock starts at bar 1, not during the lead-in. It is sent with the audio blocks, so a small ASIO buffer (128–256 samples) gives the tightest timing. RetroSeq does not receive clock.

Settings That Are Remembered

RetroSeq saves its settings when you quit: audio device and buffer, active MIDI inputs, the scanned VST3 list, metronome, lead-in, MIDI Thru, screen lines, and the outputs and instruments of all tracks. The next start opens with the same instrument setup.

12. Troubleshooting

Most problems come down to three things: no audio device open, the wrong output or channel on the track, or another program holding a MIDI port. Check them in that order.

No Sound

Symptom	Likely cause	What to do
Nothing plays, the Bar field does not move	No audio device open; the status area shows *No audio*	[F4] → *Audio and MIDI devices*, choose a device. The audio card is the sequencer's clock, even for MIDI-only songs
The position moves but one track is silent	No output, MUTE, or another track is SOLO	Check the Output and Mute columns and the SOLO indicator in the status area
External synth silent	Wrong MIDI channel, or the synth is set to another receive channel	Compare the track's **Ch** with the synth's receive channel
VST3 track silent, others play	The instrument's output pair is not the one you listen to	Check *Output* under [F4]; with a mixer interface, the pair routed to the main mix

Symptom	Likely cause	What to do
You hear a hardware synth only through its own outputs	The track plays an editor plug-in, such as a hardware synth's companion VST3, which controls the synth but makes no sound itself	Route the synth's audio through your mixer, or use the synth's MIDI port as the output
Keyboard does nothing	Input not active, or MIDI Thru OFF	[F4]: tick the keyboard under *Active MIDI inputs*. [Shift]-[F9] or OPTIONS: MIDI Thru CURRENT

Timing and Delay

Symptom	Likely cause	What to do
External MIDI arrives seconds late	Versions before 0.06 scheduled MIDI against a different Windows clock	Use 0.06 or later; the version is in the title bar
Playing a VST3 instrument feels sluggish	Large audio buffer	Lower the buffer in [F4], or switch to ASIO
Crackles with heavy instruments	Buffer too small for the CPU load	Raise the buffer; use the Release build, not Debug
Recorded notes are early or late	Monitoring latency while playing	Use a smaller buffer, or fix afterwards with the Qnt column

Notes That Hang

Symptom	What to do
A note keeps sounding after stop	Press [Alt]-P (panic): all notes off and sustain off on every output and channel
Notes hang after changing output or channel during playback	Press [Alt]-P; RetroSeq releases notes when you change a track, but some synths miss it
Sustain stays down	[Alt]-P also sends sustain pedal up

Ports and Plug-ins

Symptom	Likely cause	What to do
Could not open MIDI output … in use by another program?	Another program holds the port; older Windows MIDI drivers allow one program per port	Close the other program, or choose another port
A VST3 instrument is missing from [Ctrl]-O	Not scanned, or in a folder that is not scanned, or it is an effect	[F4] → *VST3 plug-ins*, add its folder with **+**, scan again. Only instruments are listed
The scan stays at 0 %	A large instrument loading, or a licence dialog waiting behind the window	Wait; look for a dialog in the taskbar. If RetroSeq hangs, end it and scan again: that plug-in is skipped
An instrument comes back without its sound	The `.rsq` file is missing next to the `.mid`	Keep the two files together; save again with **F** → *Save*
Program change does nothing on a VST3 track	The plug-in has no program list	Choose sounds in its own editor ([Ctrl]-E); the sound is saved in the `.rsq`

Debug Build Messages

When RetroSeq runs from Visual Studio with [F5], the debugger may stop on JUCE checks or show lines such as `ReturnHr` or `LogHr` from Windows components. These are diagnostics, not errors. Run with [Ctrl]+[F5] or use the Release build.

13. Appendix

Quick Reference Card

Screen	Keys
Everywhere	[F1] help · [F2] click · [F3] options · [F4] setup · [F5] play range · [F8] bar · [F10] quit · [Spacebar] play/stop · [Esc] back
Everywhere, Ctrl	B backup · C channel · G track · O output · P program · R record · E VST3 editor · [Alt]-P panic

Screen	Keys
Main	←→ column · + − [ ] . value · P punch-in · X XFORMS · **D**elete **L**oop **M**ute **N**ame **Q**uit **R**ecord **S**olo **T**empo · **F**ILES **O**PTIONS **V**IEW **E**DIT
View	←→ bar · [Ins] [Del] [Bksp] bars · **A**dd **C**opy **D**elete **G**oto-bar **I**nsert **R**eplace **Z**ap **W**idth · [Ctrl]-R record from cursor · B block moves · P punch-in · X XFORMS on a bar range
Edit	↑↓ pitch · ←→ unit · **L**ength **P**itch **S**tart then + − [ ] < > · [Ins] [Del] note · * copy · [Ctrl]-W/D/H/Q/E/S/T note values · K step record · M MIDI Edit · **U**nits
Mouse	click menu word = command · middle button = mouse menu · right double-click = Esc · drag up/down = value (left ±1, right ±10) · double-click = open/enter · drag notes in Edit · wheel

File Formats

File	Contents
`.mid`	Standard MIDI file type 1: tempo and time signature, one track per RetroSeq track with its name and events. Opens in any sequencer
`.rsq`	RetroSeq setup next to the `.mid`: outputs, VST3 instruments and their state, column values
`RetroSeq.settings`	Program settings in the user's application data folder

Not Yet Implemented

These commands from the classic sequencers show a message when pressed: GROUP and group arrange, H_MULTI multi-track recording, the Note Edit screen, the tempo track, sync to an external clock and chase, markers, notepad, program names, the DDL calculator and the QWERTY synth.

Other limits: one tempo per song (tempo changes in imported files are ignored), system exclusive is played back from files but not recorded, and only VST3 instruments are listed as outputs.

Version History

Version	Changes
0.16	Mouse: clickable menus, mouse menu, right double-click = Esc, drag to change values, mark bars in View, move and resize notes in Edit, wheel
0.15	Play Range ([F5]), Punch-in (P), MIDI clock out with Start/Stop/Continue and Song Position Pointer, MIDI Edit screen
0.14	Block Moves on the View screen (B); Step Record on the Edit screen (K); [Ctrl]-B undoes on several tracks at once
0.13	XFORMS: shift, scale, Super-Quantize with swing, humanize, velocity, transpose, invert, length, legato, extract and delete notes
0.12	FILES: export the current track as a MIDI file; import one part of a MIDI file into the current track
0.11	Tracks can end in empty bars: Ins or Add past the end extends the track, and a loop covers its full length
0.10	Load and New ask whether to keep the current instruments
0.09	Vol and Pan on VST3 tracks set the instrument's output level and balance directly, so they work with every plug-in
0.08	Typing replaces a prompt's suggested answer; a loop is measured to the bar of its last note start, and notes ringing past the loop end are released
0.07	Edit screen; buffers in View (Add, Copy, Delete, Zap, Insert, Replace); VST3 program selection from the Prg column and [Ctrl]-P; 25/43/50/60 screen lines
0.06	Version number on screen; external MIDI sent directly each audio block, fixing a delay of several seconds on Windows

Version	Changes
Earlier	Main and View screens with the classic menu letters, function and Ctrl keys; Prg, Trans, Qnt, Loop, Vol and Pan columns; English interface; ASIO; MIDI files and `.rsq` setups; VST3 hosting

Glossary

Term	Meaning
ASIO	A Windows audio driver standard with low latency, offered by most music interfaces
Backup	[Ctrl]-B: restores the current track as it was before the last recording or edit
Bar (measure)	One cycle of the time signature; 4/4 = four quarter notes
Buffer	One of eleven storage places (0–9, T) for bars copied in the View screen
Channel	One of 16 MIDI channels on a port; a synth listens on the channels it is set to
Click	1/480 of a quarter note, RetroSeq's finest time step; also the metronome sound
Current track	The track under the cursor; it receives MIDI in when Thru is CURRENT
CC (control change)	MIDI message for knobs, pedals, volume (CC7) and pan (CC10)
General MIDI (GM)	Standard set of 128 sounds and a drum map on channel 10
Lead-in	Bars played before recording starts, so you can find the beat
Loop	Repeating a track in whole bars for the rest of the song
Multi-timbral	A synth that plays different sounds on different channels at once

Term	Meaning
Output	Where a track's events go: a MIDI port or a VST3 instrument
Port	One MIDI output or input, each with its own 16 channels
Program change	MIDI message that selects a sound; the Prg column
Quantize	Moving notes to the nearest grid position; the Qnt column does it during playback only
Solo	Playing only one track
Thru	Passing what you play through to the current track's output
Time signature	Beats per bar and the beat's note value, e.g. 3/4
Transpose	Shifting pitch by semitones; the Trans column does it during playback only
Unit	The time step of the Edit screen cursor, from quarter notes to single clicks
Velocity	How hard a note was played, 1–127
Velocity scaling	Adding or subtracting a fixed amount from every note's velocity
VST3	Plug-in standard for software instruments; RetroSeq hosts VST3 instruments

Index

Topic	Where
Add bars	8 View Screen · Menu Commands
ASIO	3 Setting Up · Audio and MIDI Devices
Audio device	3 Setting Up; 11 Troubleshooting · No Sound
Backup ([Ctrl]-B)	2 User Interface · Ctrl Keys; 6 Tutorial
Bar position	7 Main Screen · Status Area
Buffers (Copy, Delete, Zap, Insert, Replace)	8 View Screen · Using the Buffers; 6 Tutorial · Lesson 2

Topic	Where
Building from source	3 Setting Up · Building RetroSeq
Channel	4 MIDI Basics · Channels and Ports; 7 Main Screen · Work Area
Delete a note	9 Edit Screen · Manipulating Notes
Delete a track	7 Main Screen · Menu Commands
Edit screen	9 Edit Screen; 6 Tutorial · Lesson 3
Files (load, save, new)	10 Global Functions · Files
Function keys	2 User Interface · Function Keys
General MIDI	4 MIDI Basics · General MIDI
Hanging notes, panic	11 Troubleshooting · Notes That Hang
Insert a note	9 Edit Screen · Manipulating Notes
Lead-in	10 Global Functions · Options Window
Length, Pitch, Start	9 Edit Screen · Manipulating Notes
Loop	7 Main Screen · Column Examples; 6 Tutorial · Lesson 1
Metronome	10 Global Functions · Metronome
MIDI inputs	3 Setting Up · Audio and MIDI Devices
MIDI Thru	10 Global Functions · MIDI Thru
Mute, Solo	7 Main Screen · Menu Commands
Options window	10 Global Functions · Options Window
Output ([Ctrl]-O)	3 Setting Up · Testing the System; 7 Main Screen · Work Area
Pan	7 Main Screen · Work Area and Column Examples
Program change	4 MIDI Basics; 7 Main Screen · Column Examples
Quantize (Qnt)	7 Main Screen · Column Examples
Record from bar 1	7 Main Screen · Record and Playback
Record from any bar	8 View Screen · Record and Playback

Topic	Where
Screen lines (25/43/50/60)	10 Global Functions · Options Window
Tempo	7 Main Screen · Menu Commands
Time signature	10 Global Functions · Options Window
Time units	9 Edit Screen · Time Units
Transpose (Trans)	7 Main Screen · Column Examples
Units	9 Edit Screen · Time Units
View screen	8 View Screen
Volume, velocity scaling	7 Main Screen · Column Examples
VST3 instruments, scanning	3 Setting Up · VST3 Instruments; 11 Troubleshooting
`.rsq` files	10 Global Functions · Files